PRODUCT USER MANUAL

For OMI_CLIMATE_ThSL

Issue: 4.0

Contributors : Corinne Derval, Maeva Monier

Approval Date : November 2025

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/06/15	All	Creation of the document		Renaud Dussurget, Karina von Schuckmann
2.0	2020/04/29	All	Update and addition of one product		C. Derval
3.0	2021/08/17	All	Major update		C. Derval
4.0	2025/05/12	All	Update of the document	Flora Gues	O. Le Galloudec

TABLE OF CONTENTS

RECORD TABLE **2**

GLOSSARY AND ABBREVIATIONS **4**

DATA ACCESS **5**

1) INTRODUCTION **6**

a) Summary 6

b) History of changes 6

2) DESCRIPTION OF THE PRODUCT SPECIFICATION **8**

a) General Information 8

b) Production System Description 9

c) Dataset details 9

3) FILE FORMAT **12**

4) FILES NOMENCLATURE **12**

5) REFERENCES **13**

GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
CGLOR	C-GLORS Reanalysis
GLOR	GLORYS2V4 Reanalysis
GREP	Global Reanalysis Ensemble Product
ORAS	ORAS5 Reanalysis
GLO	Global
CMEMS	Copernicus Marine Environment Monitoring Service
ThSL	Thermosteric Sea Level

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) OMI_CLIMATE_ThSL_GLOBAL : it provides time series and 2D data of Thermosteric Sea Level (ThSL) anomalies calculated relative to a 2005 to present reference period.
It contains 1D (time dimension) and 2D (latitude and longitude dimensions) ThSL information computed from different reanalysis and observation data. OMI_CLIMATE_ThSL_GLOBAL product is available for Global Ocean (60°S-60°N). It is calculated between 0-700 meter depths and 0-2000 meter depths (see QUID document for more information on the calculation). The interval between each data is a year. The ensemble mean, the spread and the single members are distributed in the corresponding NetCDF file.

OMI_CLIMATE_ThSL_GLOBAL includes three products:

- OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies which contains annual mean time series of 0-700m ThSL averaged over the global ocean (60°S-60°N).
- OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies which contains annual mean time series of 0-2000m ThSL averaged over the global ocean (60°S-60°N).
- OMI_CLIMATE_ThSL_GLOBAL_trend which contains the trend of ThSL averaged over the global ocean (60°S-60°N).

The product is yearly updated (i.e. temporal extension; see product improvements pages http://marine.copernicus.eu/services-portfolio/product-improvements/).
These data are computed using GLOR, CGLO, ORAS reanalysis (GREP [1]) and CORA5.2[2], ARMOR-3D[3] data from observations which provide temperatures.
The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.
Products OMI_CLIMATE_ThSL_GLOBAL are described in the Quality Information Document (QUID) CMEMS-OMI-QUID-GLO-ThSL.pdf (https://documentation.marine.copernicus.eu/QUID/CMEMS-OMI-QUID-GLO-ThSL.pdf).

b) History of changes

[1] GLOBAL_MULTIYEAR_PHY_ENS_001_031
[2] INSITU_GLO_PHY_TS_OA_MY_013_052
[3] MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012

October 2018	Publication of OMI_CLIMATE_ThSL_GLOBAL product (OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies)
April 2020	Update of OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies (with 2018 data) and publication of OMI_CLIMATE_ThSL_GLOBAL_trend.
August 2021	Major updates (as the whole observations data has been updated) and temporal extension with 2019 data.
May 2025	Update of OMI_CLIMATE_ThSL_GLOBAL products (with 2020, 2021, 2022 and 2023 data).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies_0_700 OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies_0_2000 OMI_CLIMATE_ ThSL_GLOBAL_trend
Geographical coverage	Global Ocean
Variables	thermosteric_change_in_mean_sea_level tendency_of_thermosteric_change_in_mean_sea_level
Available time series	2005-regularly updated (see product improvements pages User Notification Service | CMEMS)
Temporal resolution	Annual Averages
Horizontal resolution	
Integration depth (for example)	0-700m, 0-2000m
Format	NetCDF4

b) Production System Description

1. OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomolies_0_700/2000

Data extracted from GREP[4] or observations[5] provide the temperature which is used to compute the series of ThSL anomalies. The annual mean time series start on 2005 and regularly updated, each yearly mean is centred on the 30th of June. This product includes annual mean files of ThSL temporal series for Global Ocean (60°S-60°N). These averages are calculated from upper 700m depth and upper 2000m depth.

2. OMI_CLIMATE_ThSL_TREND

Data extracted from GREP or observations provide the temperature which is used to compute the ThSL anomalies and then the anomalies trend. The data coverage is from January 2005 and regularly updated. This product includes anomalies trends for Global Ocean (60°S-60°N). These averages are calculated from upper 2000m depth.

c) Dataset details

OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies_0_xxx contains the 1D annual global ThSL anomaly : from CGLO, GLOR, ORAS reanalysis, ARMOR-3D and CORA5.2 observations and the ensemble mean ThSL as well as the corresponding standard deviation.
thsl_mean_reanalyses [m] Ensemble Mean of Area Averaged Thermosteric Sea Level from reanalyses (GLORYS, CGLORS and ORAS) thermosteric_change_in_mean_sea_level **thsl _std_reanalyse** [m] Ensemble Standard Deviation of Area Averaged Thermosteric Sea Level from reanalyses (GLORYS, CGLORS and ORAS) thermosteric_change_in_mean_sea_level **thsl _glor** [m] Area Averaged Thermosteric Sea Level from GLORYS2V4 Sea Water Potential Temperature thermosteric_change_in_mean_sea_level

[4] GREP : GLOBAL_MULTIYEAR_PHY_ENS_001_031

[5] CORA5.2 : INSITU_GLO_PHY_TS_OA_MY_013_052 and ARMOR-3D : MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012

thsl _cglo [m] Area Averaged Thermosteric Sea Level from C-GLORS Sea Water Potential Temperature thermosteric_change_in_mean_sea_level **thsl _oras** [m] Area Averaged Thermosteric Sea Level from ORAS5 Sea Water Potential Temperature thermosteric_change_in_mean_sea_level **thsl _mean_observations** [m] Ensemble Mean of Area Averaged Thermosteric Sea Level from observations (CORA5.2 and ARMOR-3D) thermosteric_change_in_mean_sea_level **thsl _std_observations** [m] Ensemble Standard Deviation of Area Averaged Thermosteric Sea Level from observations (CORA5.2 and ARMOR-3D) thermosteric_change_in_mean_sea_level **thsl _cora** [m] Area Averaged Thermosteric Sea Level from CORA5.2 Sea Water Temperature thermosteric_change_in_mean_sea_level **thsl _armor** [m] Area Averaged Thermosteric Sea Level from ARMOR-3D Sea Water Temperature thermosteric_change_in_mean_sea_level **thsl _bretherton** [m] Area Averaged Thermosteric Sea Level from CORA5.2 Sea Water Temperature (non gridded data), algorithm from Von Schuckmann and Le Traon (2011) thermosteric_change_in_mean_sea_level **thsl _bretherton_error** [m] Error of Area Averaged Thermosteric Sea Level from CORA5.2 Sea Water Temperature (non gridded data), algorithm from Von Schuckmann and Le Traon (2011) thermosteric_change_in_mean_sea_level
OMI_CLIMATE_ThSL_GLOBAL_trend contains the 2D annual global ThSL trend : from CGLO, GLOR, ORAS reanalysis, ARMOR-3D and CORA5.2 observations and the ensemble mean ThSL as well as the corresponding standard deviation.
thsl_mean [m/year] Mean of Thermosteric Sea Level anomaly Trend tendency_of_thermosteric_change_in_mean_sea_level **thsl_std** [m/year] Standard deviation of Thermosteric Sea Level anomaly Trend tendency_of_thermosteric_change_in_mean_sea_level

thsl_glor [m/year]
Thermosteric Sea Level Anomaly Trend from GLORYS2V4 Sea Water Potential Temperature
tendency_of_thermosteric_change_in_mean_sea_level

thsl_oras [m/year]
Thermosteric Sea Level Anomaly Trend from ORAS5 Sea Water Potential Temperature
tendency_of_thermosteric_change_in_mean_sea_level

thsl_cora [m/year]
Thermosteric Sea Level Anomaly Trend from CORA5.2 Sea Water Temperature
tendency_of_thermosteric_change_in_mean_sea_level

thsl_armor [m/year]
Thermosteric Sea Level Anomaly Trend from ARMOR-3D Sea Water Temperature
tendency_of_thermosteric_change_in_mean_sea_level

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

5) REFERENCES

Quality Information Document

https://documentation.marine.copernicus.eu/QUID/CMEMS-OMI-QUID-GLO-ThSL.pdf